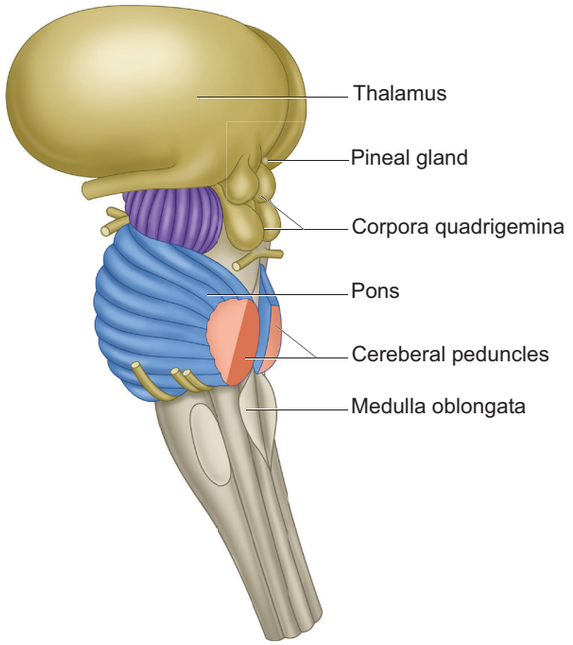



**Figure 10.10** Brain stem

### Mid brain

The mid brain is located between the diencephalon and the pons**.** The lower portion of the midbrain consists of a pair of longitudinal bands of nervous tissue called **cerebral peduncles** which relay impulses back and forth between cerebrum, cerebellum, pons and medulla. The dorsal portion of the midbrain consists of four rounded bodies called **corpora quadrigemina** which acts as a reflex centre for vision and hearing**.**

### Hind brain

Rhombencephalon forms the hind brain. It comprises of cerebellum, pons varolii and medulla oblongata. **Cerebellum** is the second largest part of the brain. It consists of two **cerebellar hemispheres** and central worm shaped part, **the vermis.** The cerebellum controls and coordinates muscular movements and body equilibrium. Any damage to cerebellum often results in uncoordinated voluntary muscle movements.

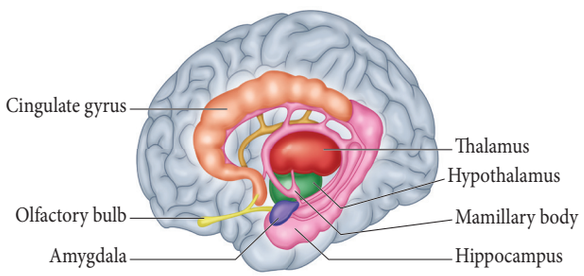



**Figure 10.9** Limbic system

**Pons varoli** lies infront of the cerebellum between the midbrain and the medulla oblongata. The nerve fibres in the pons varolii form a bridge between the two cerebellar hemispheres and connect the medulla oblongata with the other region of the brain. The respiratory nuclei found in the pons cooperate with the medulla to control respiration.

**Medulla oblongata** forms the posterior most part of the brain. It connects the spinal cord with various parts of the brain. It receives and integrates signals from spinal cord and sends it to the cerebellum and thalamus. Medulla contains vital centres that control cardio vascular reflexes, respiration and gastric secretions.

### Ventricles of the brain

The brain has four hollow, fluid filled spaces. The C- shaped space found inside each cerebral hemisphere forms the lateral **ventricles I and II** which are separated from each other by a thin membrane called the **septum pellucidum.** Each lateral ventricle communicates with the narrow III ventricle in the diencephalon through an opening called **interventricular foramen (foramen of Monro).** The ventricle III is continuous with the ventricle IV in the hind brain through a canal called **aqueduct of Sylvius (cerebral aqueduct).** Choroid plexus is a network of blood capillaries



TN_GOVT_XI_Zoology_Ch10.indd 185 09-01-2020 5.18.58 PM